PROBLEM 30

Particle on a Rotating Quarter Circumference with a Spring

Original statement · Problems.pdf, p. 18

Rotating Frame + Path Variables

A point particle P of mass m is constrained to move on a smooth quarter of circumference as shown in the figure below. We define a moving reference frame $OX_1Y_1Z_1$ that rotates with respect to an inertial reference frame $OXYZ$. The rotation of the moving reference frame is performed along a vertical axis defined as $x_1 = -R$ and $y_1 = 0$; the angular velocity is $\boldsymbol{\omega} = \omega\mathbf{k} = \omega\mathbf{k}_1$. The particle is attached to a linear spring with the other end attached to the point B of the reference frame $OX_1Y_1Z_1$ with coordinates $(-2R, 0, 0)$. The constant of the spring is $K = \alpha m\omega^2$. The angle between $\overline{OP}$ and the negative side of the z_1 axis, ϕ, is used to define the position of the particle. Neglect the action of gravity on the particle.

Initially the particle is located at $\phi = 0$ and it is set to motion with a relative velocity $-\beta\omega R\,\mathbf{i}_1$. Both α and β are positive nondimensional parameters.

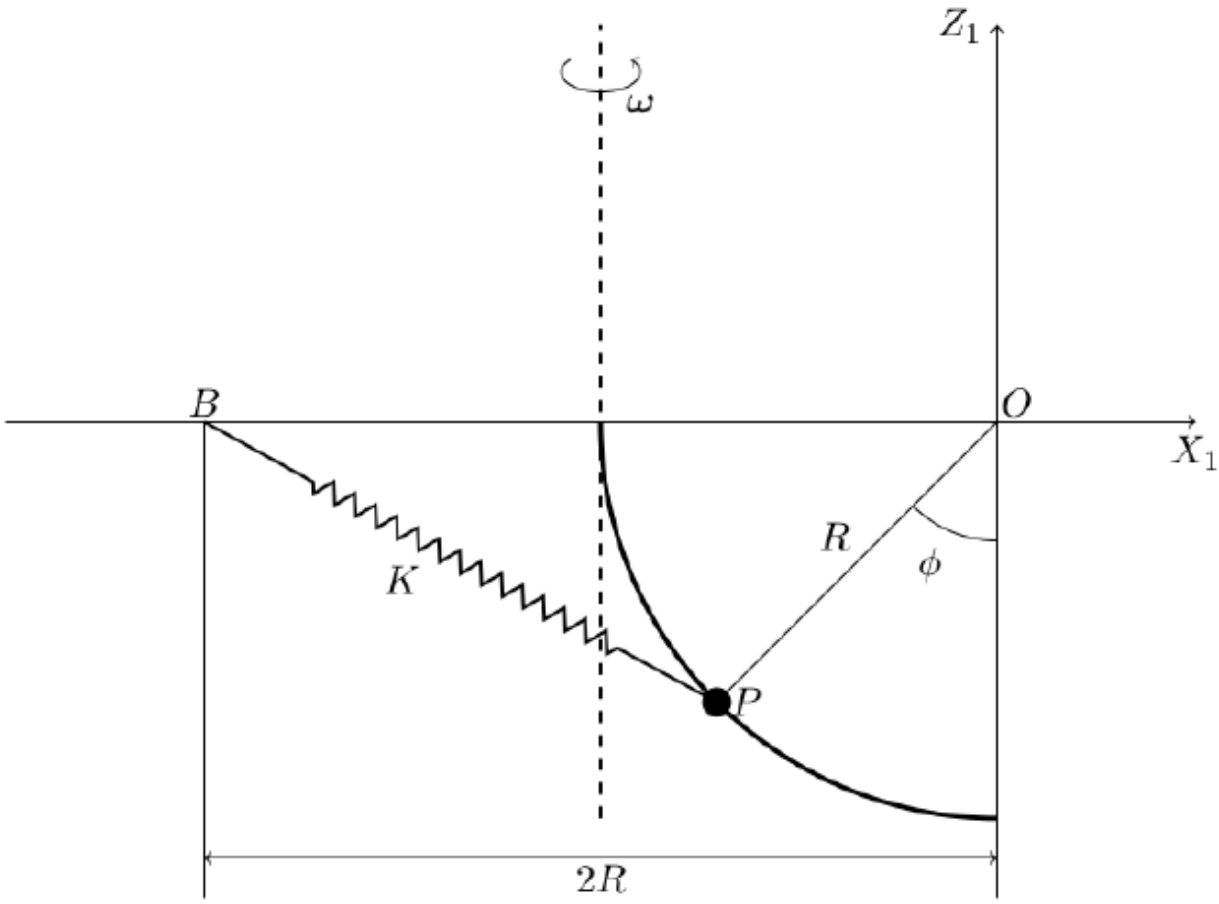


Figure: Quarter circumference of radius R centred at O in the plane X1Z1, rotation axis x1 = −R (dashed), spring of constant K from B = (−2R, 0, 0) to P, angle φ measured from the negative z1 axis (Problems.pdf, p. 18).

(a) Compute the relative velocity and the relative acceleration of the particle in the path variable reference frame $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ as functions of the coordinate ϕ and their derivatives.

(b) Write down the three components of Newton's second law expressed in the path variables reference frame $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$.

(c) Express the energy conservation equation in the relative reference frame. Discuss if this equation is equivalent to one of the above ones.

Let us now consider a different problem. Depending on the values of the parameters, we may find the equilibrium of the forces at different positions.

(d)

Obtain the value of α to have an equilibrium position when the particle is positioned at ϕ equal to $\pi/6$.

For this value of α, the particle is again located at $\phi = 0$ and is set to motion with the relative velocity defined before.

(e) Describe the possible types of motion for the different values of β. In particular, obtain the critical value of β that makes the particle reach the position $\phi = \pi/2$ and the reaction force of the curve to the particle at that instant.

(f) For a value of β higher than the critical one, the particle leaves the arc of circumference and its motion is not anymore constrained. Under these conditions compute the maximum height reached by the particle, z_{max}, as a function of β. Note that the particle is still under the action of the spring.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a particle P of mass m slides on a smooth quarter circumference of radius R and centre O, drawn in the vertical plane X_1Z_1 of a frame that rotates with constant angular velocity $\omega\mathbf{k}_1$ about the vertical axis $x_1 = -R, y_1 = 0$. A spring of constant $K = \alpha m\omega^2$ joins P to the point $B = (-2R, 0, 0)$ of the rotating frame. Gravity is neglected.
- **Why a rotating frame.** The wire and the spring anchor B are fixed in $OX_1Y_1Z_1$, so the constraint is a fixed circle there and the position of P is described by one angle. The price is the inertia force of Notes Eqs. 4.62–4.63, which is easy to compute because $\boldsymbol{\omega}_{10}$ is constant. In the inertial frame the wire itself moves, and the constraint would be harder to write.
- **Hypotheses:**
  1. $OXYZ$ is inertial; the frame rotates uniformly, $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$ with ω constant.
  2. The arc is smooth (no friction) and acts as a bilateral constraint (a bead on a wire): the reaction has no tangential component but its normal and binormal components can take

either sign.

3. The spring is linear and, since no natural length is given, of **zero natural length**: $\mathbf{F}_K = -K\,\overrightarrow{BP}$.

4. Gravity is neglected, as the statement says.

- **Degrees of freedom:** on the arc, P has one degree of freedom, the angle $\phi \in [0, \pi/2]$ between $\overline{OP}$ and the negative z_1 axis. After leaving the arc in part (f), P is free: three degrees of freedom (x_1, y_1, z_1).
- **Initial conditions:** $\phi(0) = 0$, $\mathbf{v}_1^P(0) = -\beta\omega R\,\mathbf{i}_1$.

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the arc and distance from O to the axis	m
ω	constant angular velocity of the frame	rad/s
$K = \alpha m\omega^2$	spring constant	N/m = kg/s^2
α, β	positive nondimensional parameters	1
ϕ	angle between $\overline{OP}$ and the $-z_1$ axis	rad
N_n, N_b	components of the reaction of the arc	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Fixed frame.** The only points that do not move are those of the rotation axis. We take the inertial frame $S_0 = \{Q; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with origin at the point Q of the axis at the height of O (the point $x_1 = -R$, $y_1 = z_1 = 0$, the upper end of the arc). It plays the role of the statement's $OXYZ$; moving the origin of an inertial frame to another fixed point does not change any velocity or acceleration. $\mathbf{k}_0 = \mathbf{k}$ is vertical.
- **Moving frame.** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ is the statement's $OX_1Y_1Z_1$, attached to the arc, with $\mathbf{k}_1 = \mathbf{k}_0$. Its origin O is at distance R from the axis, so it moves on a horizontal circle.
- **Change of basis.** S_1 turns about $\mathbf{k}_0$ through the angle ωt (we choose $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$). The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- **Checks.** $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$. The columns have unit length ($\cos^2\omega t + \sin^2\omega t = 1$) and the first two are orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.
- **Angular motion of S_1:** $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$ and, ω being constant, $\boldsymbol{\alpha}_{10} = \left.\dfrac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \mathbf{0}$.
- **Motion of the origin O.** $\mathbf{r}_0^O = \overrightarrow{QO} = R\,\mathbf{i}_1$. By the Coriolis theorem (Notes Eq. 3.8), since $R\,\mathbf{i}_1$ is constant in S_1:

$$\mathbf{v}_0^O = \left.\frac{d(R\mathbf{i}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times R\,\mathbf{i}_1 = \mathbf{0} + \omega R\,(\mathbf{k}_1\times\mathbf{i}_1) = \omega R\,\mathbf{j}_1$$

$$\mathbf{a}_0^O = \left.\frac{d(\omega R\mathbf{j}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\omega R\,\mathbf{j}_1 = \omega^2 R\,(\mathbf{k}_1\times\mathbf{j}_1) = -\omega^2 R\,\mathbf{i}_1$$

- **Path (intrinsic) basis.** On the arc, $\mathbf{r}_1^P = -R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$. We take the arc length $s = R\phi$ (increasing with ϕ), the tangent $\mathbf{e}_t$ in the direction of increasing ϕ, the principal normal $\mathbf{e}_n$ towards the centre of curvature O, and $\mathbf{e}_b = \mathbf{e}_t\times\mathbf{e}_n$:

$$\mathbf{e}_t = \frac{1}{R}\frac{d\mathbf{r}_1^P}{d\phi} = -\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1, \qquad \mathbf{e}_n = -\frac{\mathbf{r}_1^P}{R} = \sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{k}_1$$

$$\mathbf{e}_b = \mathbf{e}_t\times\mathbf{e}_n = \begin{vmatrix} \mathbf{i}_1 & \mathbf{j}_1 & \mathbf{k}_1 \\ -\cos\phi & 0 & \sin\phi \\ \sin\phi & 0 & \cos\phi \end{vmatrix} = (0-0)\,\mathbf{i}_1 - (-\cos^2\phi - \sin^2\phi)\,\mathbf{j}_1 + (0-0)\,\mathbf{k}_1 = \mathbf{j}_1$$

- The matrix whose columns are $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$ in $\mathcal{B}_1$ is

$$[{}_1R_t] = \begin{pmatrix} -\cos\phi & \sin\phi & 0 \\ 0 & 0 & 1 \\ \sin\phi & \cos\phi & 0 \end{pmatrix}, \qquad \det[{}_1R_t] = -\cos\phi\,(0-\cos\phi) - \sin\phi\,(0-\sin\phi) = 1$$

and its columns are unit and mutually orthogonal ($\mathbf{e}_t \cdot \mathbf{e}_n = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0$), so $[{}_1R_t][{}_1R_t]^T = I$.

- **Useful projections:** $\mathbf{i}_1 \cdot \mathbf{e}_t = -\cos\phi$, $\mathbf{i}_1 \cdot \mathbf{e}_n = \sin\phi$, $\mathbf{k}_1 \cdot \mathbf{e}_t = \sin\phi$, $\mathbf{k}_1 \cdot \mathbf{e}_n = \cos\phi$, $\mathbf{j}_1 = \mathbf{e}_b$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative velocity and acceleration in the path basis

Pedagogical Justification: The relative velocity and acceleration are the first and second derivatives of $\mathbf{r}_1^P$ taken in S_1, where $\mathbf{i}_1, \mathbf{k}_1$ are constant. Only ϕ depends on time, so each derivative is a chain rule.

Differentiate $\mathbf{r}_1^P = -R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$ with $\dfrac{d}{dt}\sin\phi = \cos\phi\,\dot\phi$ and $\dfrac{d}{dt}\cos\phi = -\sin\phi\,\dot\phi$:

$$\mathbf{v}_1^P = \left.\frac{d\mathbf{r}_1^P}{dt}\right|_1 = -R\cos\phi\,\dot\phi\,\mathbf{i}_1 + R\sin\phi\,\dot\phi\,\mathbf{k}_1 = R\dot\phi\,(-\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1)$$

$$\boxed{\mathbf{v}_1^P = R\dot\phi\,\mathbf{e}_t}$$

which is $\dot s\,\mathbf{e}_t$ with $s = R\phi$. For the acceleration use the product rule on $R\dot\phi\,\mathbf{e}_t$, and the chain rule on $\mathbf{e}_t(\phi(t))$:

$$\left.\frac{d\mathbf{e}_t}{dt}\right|_1 = \frac{d\mathbf{e}_t}{d\phi}\,\dot\phi = (\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{k}_1)\,\dot\phi = \dot\phi\,\mathbf{e}_n$$

$$\mathbf{a}_1^P = \left.\frac{d\mathbf{v}_1^P}{dt}\right|_1 = R\ddot\phi\,\mathbf{e}_t + R\dot\phi\left.\frac{d\mathbf{e}_t}{dt}\right|_1 \implies \boxed{\mathbf{a}_1^P = R\ddot\phi\,\mathbf{e}_t + R\dot\phi^2\,\mathbf{e}_n}$$

The normal part is $\dot{s}^2/\rho$ with radius of curvature $\rho = R$, as expected for a circle. There is no binormal component.

(b) Newton's second law in the path basis

> **Pedagogical Justification:** S_1 is not inertial, so we use Newton's law with inertia forces, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63, Slides 27–28). This keeps the simple relative acceleration of part (a) and puts the frame's motion into four forces.

Actual forces.

- Spring (zero natural length): $\overrightarrow{BP} = \mathbf{r}_1^P - \mathbf{r}_1^B = (2R - R\sin\phi)\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$, so

$$\mathbf{F}_K = -\alpha m\omega^2 R\,[(2 - \sin\phi)\,\mathbf{i}_1 - \cos\phi\,\mathbf{k}_1]$$

- Reaction of the smooth arc: $\mathbf{N} = N_n\,\mathbf{e}_n + N_b\,\mathbf{e}_b$ (no $\mathbf{e}_t$ component, because the arc is smooth).

Inertia forces (Slide 28), with $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{a}_0^O = -\omega^2 R\,\mathbf{i}_1$:

- Force due to the acceleration of the origin: $\mathbf{F}_A = -m\mathbf{a}_0^O = m\omega^2 R\,\mathbf{i}_1$.
- Euler force: $\mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.
- Centrifugal force. First $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \omega\mathbf{k}_1 \times (-R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1) = -\omega R\sin\phi\,\mathbf{j}_1$ (using $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$); then $\boldsymbol{\omega}_{10} \times (-\omega R\sin\phi\,\mathbf{j}_1) = -\omega^2 R\sin\phi\,(\mathbf{k}_1 \times \mathbf{j}_1) = \omega^2 R\sin\phi\,\mathbf{i}_1$, so $\mathbf{F}_C = -m\omega^2 R\sin\phi\,\mathbf{i}_1$.
- Coriolis force: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \omega R\dot{\phi}\,\mathbf{k}_1 \times (-\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1) = -\omega R\dot{\phi}\cos\phi\,\mathbf{j}_1$, so $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\omega R\dot{\phi}\cos\phi\,\mathbf{j}_1$.

The first and third combine into the centrifugal force about the **true** rotation axis, whose distance to P is $x_1 + R = R(1 - \sin\phi)$ (Slide 30):

$$\mathbf{F}_A + \mathbf{F}_C = m\omega^2 R\,(1 - \sin\phi)\,\mathbf{i}_1$$

Projections. With the projections listed in Phase 2:

$$\mathbf{F}_K \cdot \mathbf{e}_t = -\alpha m\omega^2 R\,[(2 - \sin\phi)(-\cos\phi) + (-\cos\phi)(\sin\phi)] = -\alpha m\omega^2 R\,[-2\cos\phi + \sin\phi\cos\phi - \sin\phi\cos\phi] = 2\alpha m\omega^2 R\cos\phi$$

$$\mathbf{F}_K \cdot \mathbf{e}_n = -\alpha m\omega^2 R\left[(2-\sin\phi)\sin\phi - \cos^2\phi\right] = -\alpha m\omega^2 R\left[2\sin\phi - 1\right] = \alpha m\omega^2 R\,(1-2\sin\phi)$$

$$(\mathbf{F}_A + \mathbf{F}_C)\cdot\mathbf{e}_t = -m\omega^2 R\,(1-\sin\phi)\cos\phi, \qquad (\mathbf{F}_A + \mathbf{F}_C)\cdot\mathbf{e}_n = m\omega^2 R\,(1-\sin\phi)\sin\phi$$

and $\mathbf{F}_K$, $\mathbf{F}_A + \mathbf{F}_C$ have no $\mathbf{e}_b = \mathbf{j}_1$ component, while $\mathbf{F}_{COR}$ lies entirely along $\mathbf{e}_b$.

The three components of $m\mathbf{a}_1^P = \mathbf{F}_K + \mathbf{N} + \mathbf{F}_A + \mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_{COR}$:

$$\mathbf{e}_t: \quad mR\ddot{\phi} = 2\alpha m\omega^2 R\cos\phi - m\omega^2 R\,(1-\sin\phi)\cos\phi$$

$$\mathbf{e}_n: \quad mR\dot{\phi}^2 = N_n + \alpha m\omega^2 R\,(1-2\sin\phi) + m\omega^2 R\,(1-\sin\phi)\sin\phi$$

$$\mathbf{e}_b: \quad 0 = N_b + 2m\omega R\dot{\phi}\cos\phi$$

The tangential component contains no reaction and is the **equation of motion**; the other two give the **reaction**:

$$\boxed{\ddot{\phi} = \omega^2\cos\phi\;(2\alpha - 1 + \sin\phi)}$$

$$\boxed{N_n = mR\dot{\phi}^2 - \alpha m\omega^2 R\,(1-2\sin\phi) - m\omega^2 R\,\sin\phi\,(1-\sin\phi), \qquad N_b = -2m\omega R\dot{\phi}\cos\phi}$$

The binormal reaction balances the Coriolis force: the arc has to push P sideways to keep it in the rotating plane.

(c) Energy conservation in the relative frame

> **Pedagogical Justification:** In S_1 (Slide 29) the mechanical energy E_1^P is conserved if every force that does work in S_1 has a potential. Here: $\mathbf{N}$ does no work in S_1 because the arc is fixed in S_1 and $\mathbf{N} \perp \mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t$; $\mathbf{F}_{COR} \perp \mathbf{v}_1^P$ never does work (Slide 31); $\mathbf{F}_E = \mathbf{0}$; the spring is conservative; and $\mathbf{F}_A + \mathbf{F}_C$ is the centrifugal force about an axis fixed in S_1 with constant ω, which has the potential of Slide 30.

Potentials. Spring: $V_K = \frac{1}{2}K\,|\overrightarrow{BP}|^2$ with

$$|\overrightarrow{BP}|^2 = R^2\left[(2-\sin\phi)^2 + \cos^2\phi\right] = R^2\left[4 - 4\sin\phi + \sin^2\phi + \cos^2\phi\right] = R^2\,(5 - 4\sin\phi)$$

so $V_K = \frac{1}{2}\alpha m\omega^2 R^2\,(5-4\sin\phi)$. Centrifugal (Slide 30), with $r_\perp = x_1 + R = R(1-\sin\phi)$:

$$V_C = -\tfrac{1}{2}m\omega^2 r_\perp^2 = -\tfrac{1}{2}m\omega^2 R^2\,(1-\sin\phi)^2$$

(Check: in Cartesian form $V_C = -\frac{1}{2}m\omega^2(x_1+R)^2$ and $-\partial V_C/\partial x_1 = m\omega^2(x_1+R)$, the $\mathbf{i}_1$ component of $\mathbf{F}_A + \mathbf{F}_C$.)

Energy equation:

$$\boxed{E_1^P = \tfrac{1}{2}mR^2\dot{\phi}^2 + \tfrac{1}{2}\alpha m\omega^2 R^2\,(5-4\sin\phi) - \tfrac{1}{2}m\omega^2 R^2\,(1-\sin\phi)^2 = \text{const}}$$

Initial value. At $t=0$, $\mathbf{e}_t(0) = -\mathbf{i}_1$, so $\mathbf{v}_1^P(0) = R\dot{\phi}(0)\,(-\mathbf{i}_1) = -\beta\omega R\,\mathbf{i}_1$ gives $\dot{\phi}(0) = \beta\omega$, and $E_1^P(0) = \frac{1}{2}m\omega^2R^2\,(\beta^2 + 5\alpha - 1)$. Equating and dividing by $\frac{1}{2}mR^2$:

$$\dot{\phi}^2 = \omega^2\left[\beta^2 + 5\alpha - 1 - \alpha(5-4\sin\phi) + (1-\sin\phi)^2\right] \;\Longrightarrow\; \boxed{\dot{\phi}^2 = \omega^2\left[\beta^2 + (4\alpha-2)\sin\phi + \sin^2\phi\right]}$$

Equivalence. Differentiate E_1^P in time with the chain rule ($\frac{d}{dt}\sin\phi = \cos\phi\,\dot{\phi}$, $\frac{d}{dt}(1-\sin\phi)^2 = 2(1-\sin\phi)(-\cos\phi\,\dot{\phi})$):

$$\frac{dE_1^P}{dt} = mR^2\dot\phi\ddot\phi - 2\alpha m\omega^2 R^2\cos\phi\,\dot\phi + m\omega^2R^2(1-\sin\phi)\cos\phi\,\dot\phi = mR^2\dot\phi\left[\ddot\phi - 2\alpha\omega^2\cos\phi + \omega^2(1-\sin\phi)\cos\phi\right]$$

The bracket is zero exactly when the $\mathbf{e}_t$ equation of part (b) holds. So the energy equation is **not new information**: it is the first integral of the tangential component (the $\mathbf{e}_t$ equation multiplied by $R\dot\phi$ and integrated once). The $\mathbf{e}_n$ and $\mathbf{e}_b$ components are independent of it and are needed to compute the reaction.

(d) Value of α for equilibrium at $\phi = \pi/6$

> **Pedagogical Justification:** Relative equilibrium in S_1 means $\dot\phi = 0$ and $\ddot\phi = 0$ for all time. The $\mathbf{e}_t$ equation then requires that the tangential force vanish.

$$\ddot\phi = 0 \iff \cos\phi\,(2\alpha - 1 + \sin\phi) = 0$$

The root $\cos\phi = 0$ ($\phi = \pi/2$, the particle on the axis) is an equilibrium for every α. Any other equilibrium must satisfy $\sin\phi_e = 1 - 2\alpha$. For $\phi_e = \pi/6$, $\sin\phi_e = \tfrac{1}{2}$:

$$2\alpha - 1 + \tfrac{1}{2} = 0 \implies \boxed{\alpha = \tfrac{1}{4}}$$

At that equilibrium, from part (b) with $\dot\phi = 0$: $N_b = 0$ and $N_n = -\tfrac{1}{4}m\omega^2R\,(1-1) - m\omega^2R\cdot\tfrac{1}{2}\cdot\tfrac{1}{2} = -\tfrac{1}{4}m\omega^2R$, i.e. the arc pulls P away from O with force $\tfrac{1}{4}m\omega^2R$.

(e) Types of motion for $\alpha = 1/4$, critical β and reaction at $\phi = \pi/2$

> **Pedagogical Justification:** The energy integral of part (c) is a first-order equation $\dot\phi^2 = \omega^2[\beta^2 - W(\phi)]$; motion is possible only where $W(\phi) \le \beta^2$, so the shape of W classifies the motions, as with a one-dimensional potential.

With $\alpha = \tfrac{1}{4}$, $4\alpha - 2 = -1$:

$$\dot\phi^2 = \omega^2\left[\beta^2 - W(\phi)\right], \qquad W(\phi) = \sin\phi\,(1-\sin\phi)$$

On $[0,\pi/2]$: $W(0)=0$, $W(\pi/2)=0$, and $\dfrac{dW}{d\phi}=\cos\phi\,(1-2\sin\phi)$ vanishes inside the interval only at $\sin\phi=\tfrac12$, i.e. $\phi=\pi/6$, where W has its maximum $W(\pi/6)=\tfrac12\cdot\tfrac12=\tfrac14$. W plays the role of a potential barrier, so the equilibrium $\phi=\pi/6$ is **unstable**. Three types of motion follow:

- $\beta>\tfrac12$**:** $\dot\phi^2\geq\omega^2(\beta^2-\tfrac14)>0$ on the whole arc, so ϕ increases monotonically, crosses the barrier at $\pi/6$ and reaches $\phi=\pi/2$ with $\dot\phi^2=\omega^2[\beta^2-0]$, i.e. $\dot\phi=\beta\omega$: the relative speed there equals the initial one, $\beta\omega R$. The particle then leaves the arc at its upper end (part f).
- $\beta=\tfrac12$**:** $\beta^2-W=\tfrac14-\sin\phi+\sin^2\phi=(\tfrac12-\sin\phi)^2$, so $\dot\phi=\omega(\tfrac12-\sin\phi)$. The particle approaches the unstable equilibrium $\phi=\pi/6$ asymptotically and never reaches it: near it, $\sin\phi\approx\tfrac12+\tfrac{\sqrt3}{2}(\phi-\tfrac{\pi}{6})$ and $\dot\phi\approx-\tfrac{\sqrt3}{2}\omega\,(\phi-\tfrac{\pi}{6})$, an exponential approach with rate $\tfrac{\sqrt3}{2}\omega$.
- $\beta<\tfrac12$**:** $\dot\phi$ vanishes at the turning point ϕ^* with $W(\phi^*)=\beta^2$, i.e. $\sin^2\phi^*-\sin\phi^*+\beta^2=0$; the root reached first is

$$\sin\phi^*=\frac{1-\sqrt{1-4\beta^2}}{2}<\frac12$$

There $\ddot\phi=\omega^2\cos\phi^*\,(\sin\phi^*-\tfrac12)<0$, so the particle turns back, returns to $\phi=0$ with $\dot\phi=-\beta\omega$ (relative velocity $+\beta\omega R\,\mathbf{i}_1$, same speed by energy conservation) and leaves the quarter circumference at its lower end.

Critical value. The particle reaches $\phi=\pi/2$ if and only if β exceeds

$$\boxed{\beta_c=\tfrac12}$$

Reaction at $\phi=\pi/2$. Here $\cos\phi=0$, $\sin\phi=1$, $\dot\phi=\beta\omega$, and $\mathbf{e}_n(\pi/2)=\mathbf{i}_1$. From part (b) with $\alpha=\tfrac14$:

$$N_b=-2m\omega R\,\beta\omega\cdot0=0,\qquad N_n=mR\beta^2\omega^2-\tfrac14 m\omega^2R\,(1-2)-m\omega^2R\cdot1\cdot0=m\omega^2R\left(\beta^2+\tfrac14\right)$$

$$\boxed{\mathbf{N}(\phi=\tfrac{\pi}{2})=m\omega^2R\left(\beta^2+\tfrac14\right)\mathbf{i}_1\ \xrightarrow{\beta\to\beta_c}\ \tfrac12 m\omega^2R\,\mathbf{i}_1}$$

It points towards O: it supplies the centripetal force $mR\dot{\phi}^2$ and also balances the spring, which pulls P towards B with $\frac{1}{4}m\omega^2 R$. The centrifugal and Coriolis forces vanish there because P is on the axis and moves parallel to it.

(f) Maximum height after leaving the arc ($\beta > \beta_c$)

> **Pedagogical Justification:** Once free, P has three degrees of freedom, and the horizontal motion is coupled through the Coriolis and centrifugal forces. But every inertia force of a frame rotating about the vertical is horizontal, so the vertical component of $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ involves only the spring and decouples.

Initial state of the free motion (time $\tau = 0$ at separation): $\phi = \pi/2$ gives $\mathbf{r}_1^P = -R\,\mathbf{i}_1$ (on the axis, $z_1 = 0$), and $\mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t(\pi/2) = \beta\omega R\,\mathbf{k}_1$ (vertical, upward).

Vertical equation. For a free position $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1$:

- $\mathbf{k}_1 \cdot \mathbf{F}_K = -K\,\mathbf{k}_1 \cdot [(x_1 + 2R)\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1] = -\alpha m\omega^2 z_1$;
- $\mathbf{k}_1 \cdot (\mathbf{F}_A + \mathbf{F}_C) = 0$, because $\mathbf{F}_A + \mathbf{F}_C = m\omega^2\,[(x_1 + R)\mathbf{i}_1 + y_1\mathbf{j}_1]$ is horizontal;
- $\mathbf{k}_1 \cdot \mathbf{F}_{COR} = -2m\,\mathbf{k}_1 \cdot (\omega\mathbf{k}_1 \times \mathbf{v}_1^P) = 0$, because $\mathbf{k}_1 \times \mathbf{v}_1^P \perp \mathbf{k}_1$.

Hence, with $\alpha = \frac{1}{4}$,

$$m\ddot{z}_1 = -\alpha m\omega^2 z_1 \implies \ddot{z}_1 + \frac{\omega^2}{4} z_1 = 0$$

a harmonic oscillator of angular frequency $\omega/2$. Its general solution and derivative (chain rule $\frac{d}{d\tau}\sin\frac{\omega\tau}{2} = \frac{\omega}{2}\cos\frac{\omega\tau}{2}$) are

$$z_1 = C_1\cos\frac{\omega\tau}{2} + C_2\sin\frac{\omega\tau}{2}, \qquad \dot{z}_1 = -\frac{\omega}{2}C_1\sin\frac{\omega\tau}{2} + \frac{\omega}{2}C_2\cos\frac{\omega\tau}{2}$$

Initial conditions: $z_1(0) = C_1 = 0$ and $\dot{z}_1(0) = \frac{\omega}{2}C_2 = \beta\omega R$, so $C_2 = 2\beta R$:

$$z_1(\tau) = 2\beta R\,\sin\frac{\omega\tau}{2} \implies \boxed{z_{max} = 2\beta R} \quad \text{reached at } \tau = \pi/\omega$$

Since $\mathbf{k}_1 = \mathbf{k}$, this is also the height in the inertial frame, measured from the horizontal plane through O and B. For a general α the same steps give $z_{max} = \beta R/\sqrt{\alpha}$. (Equivalently, the vertical motion conserves $\frac{1}{2}m\dot{z}_1^2 + \frac{1}{2}Kz_1^2$, and $\frac{1}{2}m(\beta\omega R)^2 = \frac{1}{2}\cdot\frac{1}{4}m\omega^2 z_{max}^2$ gives the same

result.) The result assumes that the free particle does not hit the wire again.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Competition of two forces.** Along the arc the spring pulls P towards B (towards larger ϕ) with tangential force $2\alpha m\omega^2 R\cos\phi$, while the centrifugal force about the axis pushes it away from the axis (towards smaller ϕ) with $m\omega^2 R(1-\sin\phi)\cos\phi$. Both scale with ω^2 because $K=\alpha m\omega^2$, so ω only sets the time scale $1/\omega$ and the dynamics depends on α and β alone.
- **Limiting cases.** For $\alpha>\frac{1}{2}$ the factor $2\alpha-1+\sin\phi$ is positive on the whole arc and there is no interior equilibrium. For $\beta\to\beta_c^+$ the time to cross the barrier grows without bound (logarithmically), consistent with the asymptotic motion at $\beta=\beta_c$. As $\phi\to\pi/2$, both the Coriolis force and N_b vanish, as they must for motion parallel to $\boldsymbol{\omega}_{10}$.
- **Inertial check.** Computing $\mathbf{a}_0^P$ by differentiating $\mathbf{r}_0^P=\mathbf{r}_0^O+[{}_0R_1]\mathbf{r}_1^P$ twice gives exactly $\mathbf{a}_0^O+\mathbf{a}_1^P+\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P)+2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27), and $m\mathbf{a}_0^P=\mathbf{F}_K+\mathbf{N}$ holds with the reactions of part (b) (checked symbolically with sympy).
- **Dimensions.** $[K]=[\alpha m\omega^2]=\mathrm{kg}\cdot\mathrm{s}^{-2}=\mathrm{N/m}$; $[\ddot{\phi}]=[\omega^2]=\mathrm{s}^{-2}$; $[N_n]=[m\omega^2 R]=\mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2}=\mathrm{N}$; $[E_1^P]=[m\omega^2R^2]=\mathrm{J}$; $[z_{max}]=[\beta R]=\mathrm{m}$.

Source: Problems.pdf (PDF page 18)